

Lecture #31,32

Content to be covered

- ❖ Planck's Quantum Postulates
- ❖ Average energy of Planck's oscillator
 - Deduction of Rayleigh-Jean's Law and Wien's Law
- ❖ Wien's displacement law
- ❖ Stefan-Boltzmann Law
- ❖ Summary of Black Body radiation

Resources to be consulted

- Concepts of Modern Physics by Arthur Beiser

RI L $\rightarrow$ standing wave larger λ lower ν

$U(\nu) d\nu = \frac{8\pi \nu^2}{c^3} d\nu \times (kT)$

WL gas molecule $E(\nu) d\nu = \frac{8\pi \nu^2}{c^3} d\nu \times h\nu e^{-h\nu/kT}$

PR L $= \frac{2h\nu}{h\nu} \rightarrow \frac{2h\nu}{h\nu} \times \frac{8\pi \nu^2}{c^3} d\nu \times \frac{h\nu}{e^{h\nu/kT} - 1}$

Planck's Quantum Postulates

1. A black body radiation chamber is filled up not only with radiation, but also with simple harmonic oscillators or resonators (energy emitters) of the molecular dimensions, known as Planck's oscillators or Planck's resonators, which can vibrate, with all possible frequencies. **The vibration of the resonator entails one degree of freedom only.**
2. The oscillators (or resonators) cannot radiate or absorb energy continuously, but energy is emitted or absorbed in the form of packets or quanta called photons. the theory states that the exchange of energy between radiation and matter cannot take place continuously but only in certain multiples of the fundamental frequency of the **resonator (energy emitter)**.

*As the energy of a photon is $h\nu$, the energy emitted (or absorbed) is equal to $0, h\nu, 2h\nu, 3h\nu, \dots, nh\nu$, i.e., in multiplets of some small unit, called as **quantum**.*

Max Planck:

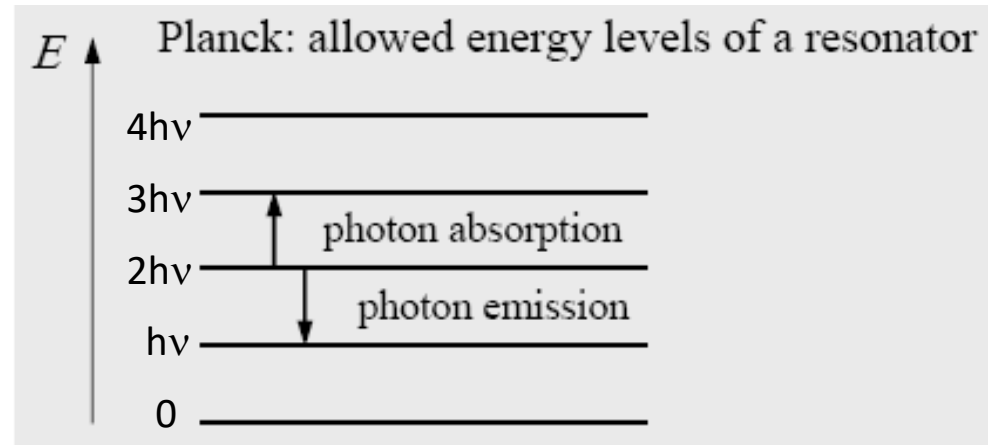
blackbody radiation is produced by vibrating submicroscopic electric charges, which he called resonators the walls of a cavity are composed of resonators vibrating at different frequency.

Classical Maxwell theory:

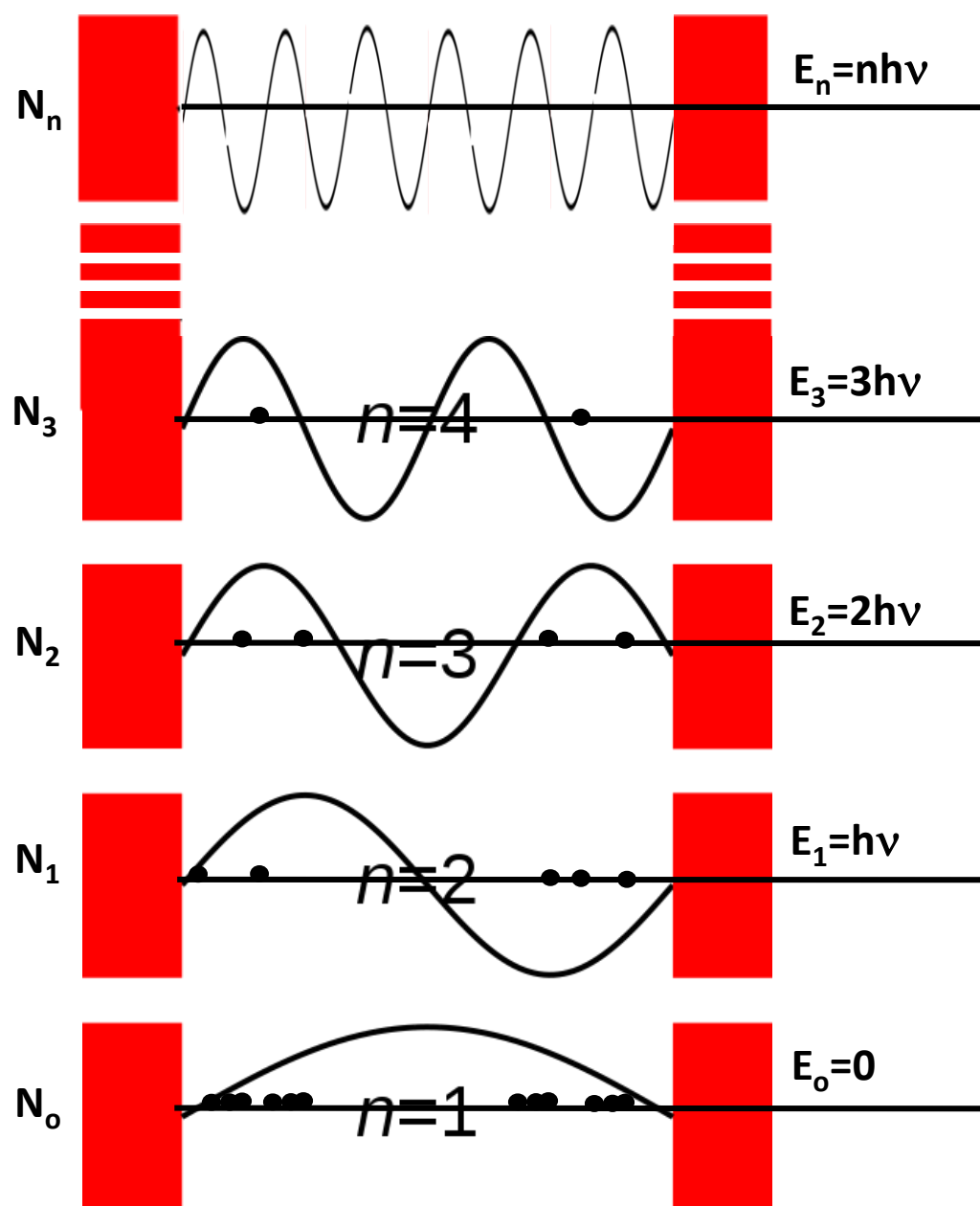
An oscillator of frequency ν could have any value of energy and could change its amplitude continuously by radiating any fraction of its energy

Planck: the total energy of a resonator with frequency ν could only be an integer multiple of $h\nu$. (During emission or absorption of light) resonator can change its energy only by the quantum of energy $\Delta E = h\nu$

$$E = nh\nu \text{ where } n = 1, 2, 3, \dots$$
$$\Delta E = h\nu$$



Total number of resonators
 $(N) = N_0 + N_1 + N_2 + N_3 + \dots + N_n$



$$\text{Total Energy (E)} = N_0 E_0 + N_1 E_1 + N_2 E_2 + N_3 E_3 + \dots + N_n E_n$$

Therefore average energy(ε) = $\frac{\sum E_n}{\sum N_n}$

Therefore, the energy density belonging to the range dv can be obtained by multiplying the average energy of Planck's oscillator by the number of oscillators per unit volume, in this frequency range ν and $(\nu + dv)$.

$$u(\nu)dv = \text{average Energy} \times \text{Number density}$$

$$= \bar{\varepsilon} \cdot n(\nu)dv$$

$$u(\nu)dv = \frac{8\pi\nu^2}{c^3} \frac{h\nu}{(e^{h\nu/kT} - 1)} dv$$

where $u(\nu)dv$ is energy density (i.e., total energy per unit volume) belonging to the range dv called Planck's radiation law in terms of frequency.

"Blackbody radiation" or "cavity radiation" refers to an object or system which absorbs all radiation incident upon it and re-radiates energy which is characteristic of this radiating system only, not dependent upon the type of radiation which is incident upon it. The radiated energy can be considered to be produced by standing wave or resonant modes of the cavity which is radiating

	Number of modes per unit frequency per unit volume	Probability of occupying modes	Average energy per mode
Classical	$8\pi\nu^2 / c^3$	Equal for all modes	kT
Quantum	$8\pi\nu^2 / c^3$	Quantized modes: require $h\nu$ energy to excite upper modes, less probable	$\frac{h\nu}{e^{h\nu/kT} - 1}$

Case I : for low frequency region; $h\nu \ll kT$

$$\Rightarrow e^{h\nu/kT} \ll 1, \quad e^{h\nu/kT} - 1 = 1 + \frac{h\nu}{kT} - 1 = \frac{h\nu}{kT}$$

$$u(\nu)d\nu = \frac{8\pi\nu^2}{c^3} \frac{h\nu}{(e^{h\nu/kT} - 1)} d\nu$$

$$\text{Hence } E(\nu)d\nu = \frac{8\pi\nu^2}{c^3} \left(\frac{h\nu}{kT} \right) d\nu = \frac{8\pi\nu^2}{c^3} kT d\nu, \quad \text{"Rayleigh Jeans Law"}$$

Case II : for high frequency region;

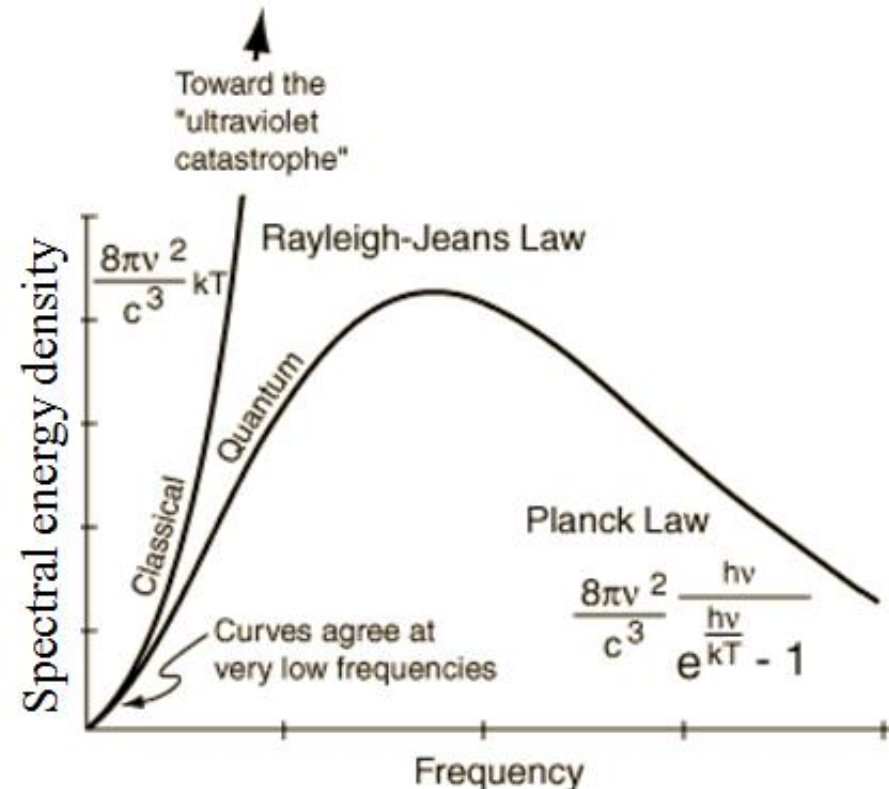
$$h\nu \gg kT, \quad e^{h\nu/kT} \gg 1,$$

$$e^{h\nu/kT} - 1 = e^{h\nu/kT}$$

$$\text{Hence, } E(\nu)d\nu = \frac{8\pi\nu^2}{c^3} \frac{h\nu}{e^{h\nu/kT} - 1} d\nu$$

$$E(\nu)d\nu = \frac{8\pi h\nu^3}{c^3} e^{-h\nu/kT} d\nu$$

known as **Wien's law**



Wien's displacement law

The wavelength whose energy density is greatest can be obtained by setting

$$\frac{du(\lambda)}{d\lambda} = 0 \quad \text{and solving for } \lambda = \lambda_{\max}. \text{ We find}$$

$$\frac{hc}{kT\lambda_{\max}} = 4.965$$

which may be written as

$$\lambda_{\max} T = \frac{hc}{4.965k}$$

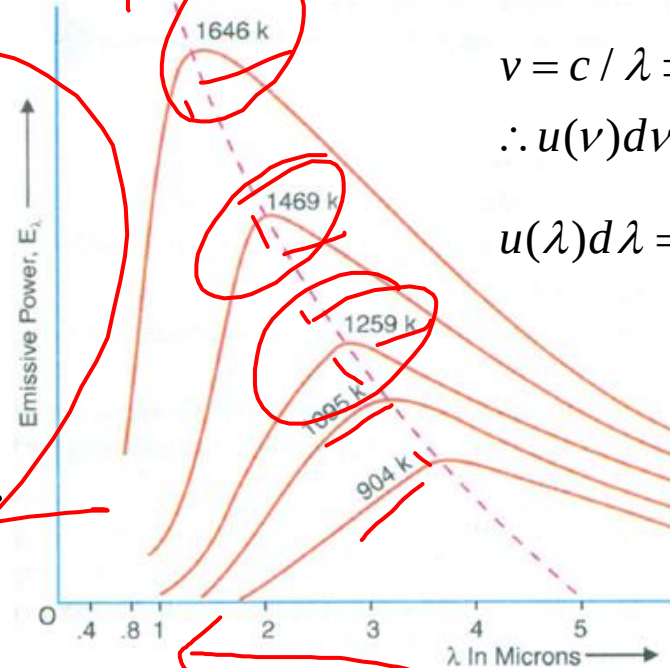
$$\lambda_{\max} T = 2.898 \times 10^{-3} \text{ m} - \text{K} = \text{const.}$$

$$u(\nu)d\nu = \frac{8\pi\nu^2}{c^3} \frac{h\nu}{(e^{h\nu/kT} - 1)} d\nu$$

$$\nu = c / \lambda \Rightarrow d\nu = -c / \lambda^2 d\lambda$$

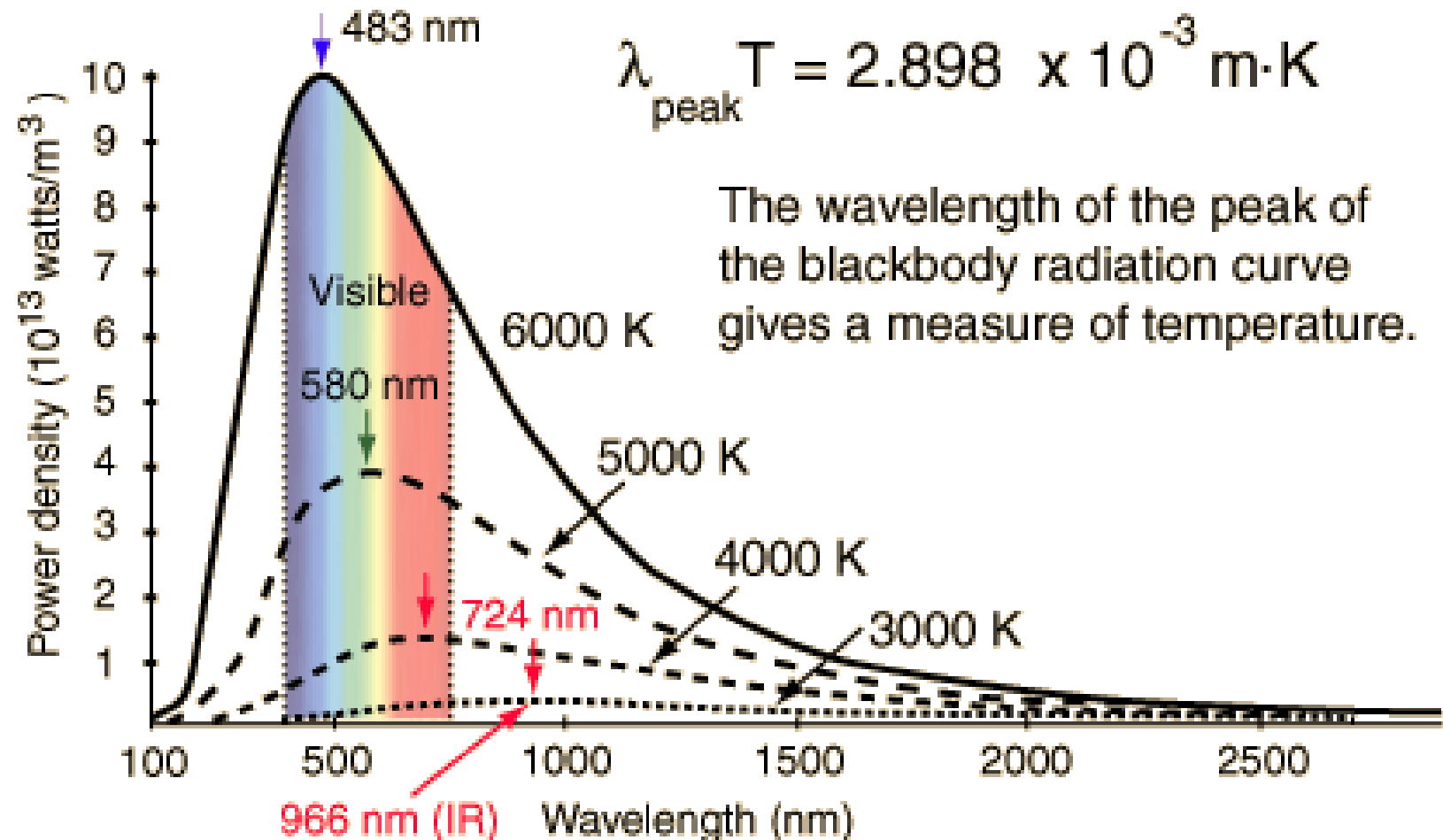
$$\therefore u(\nu)d\nu = -u(\lambda)d\lambda$$

$$u(\lambda)d\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda KT} - 1} d\lambda$$



This equation is known as Wien's displacement law. It quantitatively expresses the empirical fact that the peak in the black body spectrum shifts progressively to shorter wavelength (higher frequencies) as the temperature is increased.

Wien's displacement law



Stefan-Boltzmann Law

Total energy density within the cavity can be obtained by integrating energy density (eq. 3) over all frequencies. Thus

$$u = \int_0^{\infty} u(\nu) d\nu = \frac{8\pi^5 k^4}{15c^3 h^3} T^4$$
$$u = aT^4$$
$$u(\nu)d\nu = \frac{8\pi h}{c^3} \frac{\nu^3}{e^{h\nu/kT} - 1} d\nu$$
$$\int_0^{\infty} \frac{x^3}{e^x - 1} dx = \frac{\pi^4}{15}$$

Here ***a*** is a universal constant. According to the equation, the total energy density is proportional to the fourth power of the absolute temperature of the cavity walls. Therefore we expect that energy *E* radiated by a black body per second per unit area is also proportional to T^4 i.e.

$$E = e\sigma T^4$$

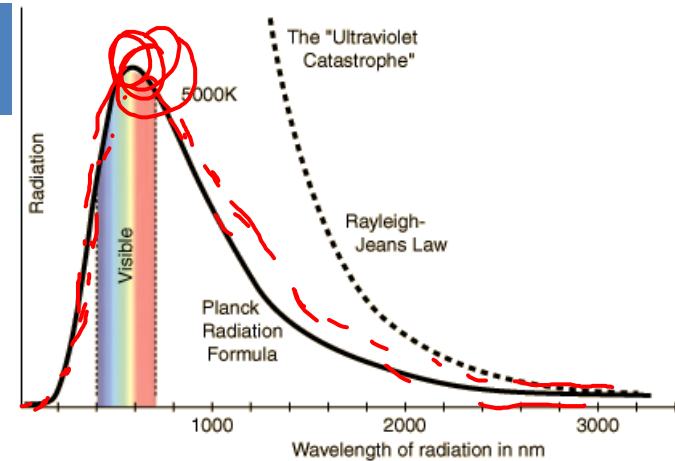
Where, *e* = emissivity, ranges from 0 (for perfectly reflecting surface) to 1 (for a black body), and σ (Stefan's constant) = $ac/4 = 5.67 \times 10^{-8} \text{ watt/m}^2\text{-K}^4$.

This equation is known as Stefan-Boltzmann Law.

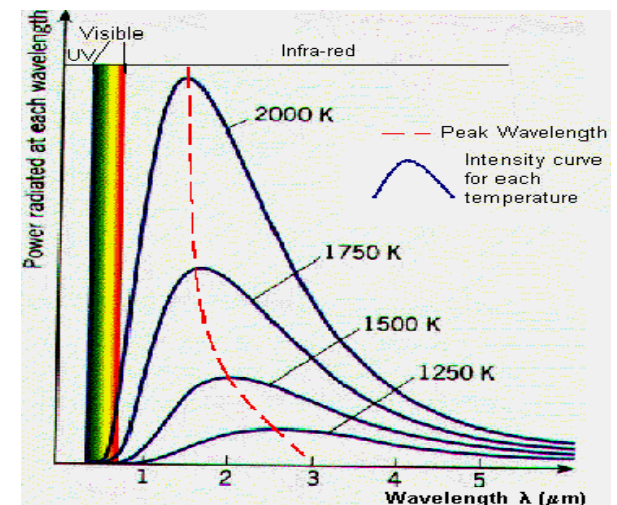
The Black Body radiation curve

1. The black body does radiate energy at every wavelength. Energy of the radiation initially increases, reaching at a maxima (peak wavelength) then decreases but never goes to zero.
2. At 5000K the peak wavelength is about (500nm or 0.5 μm) which is in the visible light region. (Example : Sun)
3. At a given T, the intensity of heat radiation increases with $\uparrow$ in λ and at a particular λ its value is maximum i.e. λ_m . With further $\uparrow$ in λ , the intensity of heat radiation $\downarrow$.
4. An $\uparrow$ in T causes an $\uparrow$ in the energy emission, for all wavelength.
5. An $\uparrow$ in T causes a $\downarrow$ in λ_m , such that $\lambda_m T = \text{constant}$

This is called Wein's displacement law. It therefore begins to move from the infra-red towards the visible part of the spectrum.



Black body radiation curve for 5000K. Peak wavelength is 500nm.



Black body radiation curves showing peak wavelengths (Red line) at various temperatures

The Black Body radiation curve

5. At each temperature the black body emits a standard amount of energy.

The area under each curve represents the total energy emitted by the body at a particular temperature for the range of λ s considered. This area $\uparrow$ s with $\uparrow$ in temperature of the body. It is found that the area is directly proportional to the fourth power of the temperature of the body, i.e., $E \propto T^4$ **this represents Stefan-Boltzmann's law.**

